

MATH 3260 - Chapter 9 Worksheets

Linear Programming

Name: _____

Date: _____

These are student practice worksheets. They are designed to be used with the chapter notebook and notes, but every problem below contains the data needed to work it independently. No solutions are included in this student version.

Section 9.1 - Systems of Linear Inequalities

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

Graphical problems in this chapter are in $\mathbb{R}^2$. Remember: multiplying or dividing an inequality by a negative number reverses the inequality sign.

1. Test $(2, 1)$, $(0, 0)$, and $(-1, -4)$ in

$$2x - y > 1.$$

2. Graph $3x + 2y \leq 12$. Show the intercepts, boundary type, a test point, and the shaded half-plane.

3. Find all vertices of

$$x \geq 0, \quad y \geq 0, \quad x + y \leq 6, \quad 2x + y \leq 8.$$

Identify the active constraints at $(2, 4)$.

4. Determine whether

$$x \geq 0, \quad y \geq 0, \quad x - y \geq 1$$

defines a bounded feasible region. Give one feasible direction if it is unbounded.

5. A fictional weekly activity plan uses x long hiking sessions and y badminton sessions. Model all constraints:

$$x + y \leq 8, \quad 2x + y \leq 12, \quad x + 2y \leq 14,$$

with $x, y \geq 0$, and add the new requirement that at least two badminton sessions are scheduled. Graph the feasible region and list its vertices.

6. Prove that the intersection of half-planes is convex.

7. Let $L(x, y) = ax + by - c$. Explain why one test point determines which side of the boundary $L(x, y) = 0$ satisfies $L(x, y) \leq 0$.

Section 9.2 - Linear Programming in Two Variables

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.
For

$$Z = 50x + 40y$$

subject to

$$x, y \geq 0, \quad x + y \leq 10, \quad 2x + y \leq 16, \quad x + 3y \leq 21,$$

complete Problems 1-3.

1. Find all feasible vertices.
2. Determine the maximum value of Z and report all slacks at an optimizer.
3. Replace the objective by $Q = 2x + y$ and identify every optimizer.
4. Solve the fictional event model

$$R = 80x + 60y$$

subject to

$$x + y \leq 9, \quad 2x + y \leq 14, \quad x + 2y \leq 16, \quad x, y \geq 0.$$

5. Diagnose unboundedness for maximizing $x + y$ subject to

$$x - y \geq 1, \quad x, y \geq 0.$$

6. Prove the vertex principle for a bounded polygonal feasible region in $\mathbb{R}^2$.

Section 9.3 - The Simplex Method: Maximization

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.
Use exact simplex arithmetic for

$$\max z = 3x_1 + 2x_2$$

subject to

$$x_1 \leq 4, \quad x_2 \leq 5, \quad x_1 + x_2 \leq 6, \quad x_1, x_2 \geq 0.$$

1. Form the equations with slack variables and write the initial tableau.
2. Show every ratio, pivot choice, normalization, and row operation.
3. After each pivot, identify the corresponding basic feasible solution. Describe the path between vertices.
4. Report the optimizer, objective value, and all slacks.
5. Prove optimality from the final objective row.
6. Explain the meaning of each exceptional signal: zero reduced cost; an entering column with no positive constraint entry; and degeneracy.

Section 9.4 - Minimization and Duality

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation. For

$$\min w = 5x_1 + 4x_2$$

subject to

$$x_1 + x_2 \geq 5, \quad 2x_1 + x_2 \geq 8, \quad x_1, x_2 \geq 0,$$

complete Problems 1-4.

1. Construct the dual and solve it, showing the simplex tableaus.
2. Recover the primal optimizer from the dual solution and verify it in the original primal constraints.
3. Check complementary slackness explicitly.
4. Prove weak duality for the primal-dual pair and explain why equality of feasible objective values is an optimality certificate.
5. Construct the dual of

$$\min 7x_1 + 5x_2$$

subject to

$$x_1 + x_2 \geq 6, \quad 3x_1 + x_2 \geq 10, \quad x_1, x_2 \geq 0.$$

6. Interpret the dual variables as shadow prices in words.

Section 9.5 - Mixed Constraints and Two-Phase Simplex

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation. Use two-phase simplex for

$$\max z = 4x_1 + 3x_2$$

subject to

$$x_1 + x_2 \leq 5, \quad 2x_1 + x_2 \geq 4, \quad x_1, x_2 \geq 0.$$

1. Introduce the needed slack, surplus, and artificial variables.
2. Complete Phase I and explain how the final Phase I objective proves feasibility.
3. Restore the original objective and complete Phase II.
4. Verify the optimizer, slack, and surplus in the original constraints.
5. Classify the added variables needed for $\leq$, $\geq$, and equality constraints.
6. Prove the Phase I criterion: optimum 0 means the original constraints are feasible, while a positive optimum signals infeasibility.
7. Convert a mixed minimization objective to a maximization objective by negation and explain what happens to the optimal value.
8. A distributor ships from Depot 1 and Depot 2 to Center 1 and Center 2. Supplies are 70 and 50 units; minimum demands are 40 and 60 units. Shipping costs are

$$\begin{bmatrix} 4 & 7 \\ 6 & 3 \end{bmatrix}.$$

Let x_{ij} be the amount shipped from Depot i to Center j . Formulate a linear program that minimizes shipping cost. Identify which constraints are supply limits and which are demand requirements.

Common mistakes: forgetting to reverse an inequality after multiplying by -1 ; ignoring variable bounds; optimizing before establishing feasibility; or applying the simplex ratio test to a nonpositive pivot-column entry.